Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/26/2021
PI Name:	Limoli, Charles Ph.D.		
Project Title:	NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-NSCOR Radiation
Start Date:	07/01/2015	**End Date:**	02/28/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2022 per NSSC information (Ed., 3/12/21) NOTE: End date changed to 2/28/2021 per NSSC information (Ed., 4/6/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Acharya, Munjal Ph.D. (University of California, Irvine) Baulch, Janet Ph.D. (University of California, Irvine) Britten, Richard Ph.D. (Eastern Virginia Medical School) Parihar, Vipan Ph.D. (University of California, Irvine) Soltesz, Ivan Ph.D. (University of California, Irvine)		
Grant/Contract No.:	NNX15AI22G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources.
Task Progress:	We have made progress on several fronts over the last, rather unusual year. Highlights of these efforts are detailed below: We have published two manuscripts detailing the adverse impact of chronic low dose rate neutron exposure on CNS functionality (Ed. note 4/27/21--both publications reported previously in FY2020 report--see Cumulative Bibliography link): 1) Acharya, M.M., Baulch, J.E., Klein, P.M., Baddour, A.A.D., Apodaca, L.A., Kramár, E.A., Alikhani, L., Garcia Jr. C., Angulo, M.C., Batra, R.S., Fallgren, C.M., Borak, T.B., Stark, C.E.L., Wood, M.A., Britten, R.A., Soltesz, I. and Limoli, C.L. New concerns for neurocognitive function during deep space exposures to chronic, low dose rate, neutron radiation. eNeuro, 22;6(4) (2019). pii: ENEURO.0094-19, doi: 10.1523/ENEURO.0094-19. 2) Britten, R.A., Duncan, V.D., Fesshaye, A., Wellman, L., Fallgren, C.M. and Sanford, L.D. Sleep fragmentation exacerbates executive function impairments induced by protracted low dose rate neutron exposure. Int. J. Radiat Biol. (2019), Dec 6:1-11. doi:10.1080/09553002.2019.1694190. A follow up study has since been published detailing similarities between the mouse and rat response to chronic neutron irradiation: 3) Krishnan, B., Natarajan, C., Bourne, K.Z., Alikhani, L., Wang, J., Sowa, A., Groen, K., Perry, B., Dickstein, D., Baulch, J.E., Limoli, C.L. and Britten, R.A. Chronic low dose neutron exposure results in altered neurotransmission properties of the hippocampus-prefrontal cortex axis in both mice and rats. Int. J. Molec. Sci. (2020), 22:3668. doi.org/=10.3390/ijms22073668. In sum, low dose and low dose rates have been found to have a significant adverse impact on CNS functionality, as evidenced by decrements at the electrophysiological and neurocognitive levels. We have since followed up those studies with an experiment implementing the same isodose of neutrons, but delivered acutely (hrs) instead of over the course of 6 months. These studies are currently ongoing, and while preliminary data indicates several of the same cognitive decrements are found after acute neutron exposures, previous decreases in long-term potentiation (LTP) found after chronic exposures were not found after acute exposures. A manuscript has now been published detailing the differences between the sexes regarding cosmic radiation exposure: 4) Parihar, V.K., Angulo, M.C., Allen, B.D., Syage, A., Usmani, M.T., Chapelle, E.P. d-l., Amin, A.N., Flores, L., Lin, X., Giedzinski, E. and Limoli, C.L. Sex Specific Cognitive Deficits Following Space Radiation Exposure. Front. Behav. Neurosci. (2020). 14:535886. doi 10.3389/fnbeh.2020.535885. Here we find that female mice are more resistant to radiation-induced cognitive decrements than male mice, a finding that is in part related to differences in inflammatory signaling between the sexes. We have also published 2 manuscripts detailing the adverse effects of the 5-6 ion Galactic Cosmic Ray (GCR) simulation: 5) Klein, P.M., Parihar, V.K., Szabo, G.G., Zöldi, M., Angulo, M.C., Allen, B.D., Amin, A.N., Nguyen, Q-A., Katona, I., Baulch, J.E., Limoli, C.L. and Soltesz, I. Detrimental impacts of mixed-ion radiation on nervous system function, Neurobiol. Dis. 151:105252 (2021), doi: 10.1016/j.nbd.2021.105252. 6) Keiser, A.A., Kramár, E.A., Dong T., Shanur S., Pirodan M., Ru N., Acharya M.M., Baulch J.E., Limoli C.L. and Wood MA. Systemic HDAC3 inhibition ameliorates impairments in synaptic plasticity caused by simulated galactic cosmic radiation exposure in male mice. Neurobiol. Learn. & Memory. (2020). 178:107367, doi: 10.1016/j.nlm.2020.107367. In the first paper, extensive electrophysiological data finds mixed ion exposure to compromise intrinsic cell properties that can be related to specific neurocognitive decrements in the irradiated rodent brain. In the second paper, mice subjected to a similar irradiation paradigm were treated with a drug known to alter the status of histone acetylation. The deacetylation inhibitor was found to resolve radiation-induced decrements at the electrophysiological level, pointing to a potentially new class of drugs that hold promise for mitigating the advert effects of space radiation exposure on CNS

functionality.

Follow up work on our extensive complex 33-beam GCR sim continues, and is summarized by bullet point below.

Behavioral testing:

*Limoli Lab: (Object Updating Learning (OUL), Novel Object Recognition (NOR), Light/Dark Box (LDB), Social Interaction Test (SIT), Fear Extinction (FE)) - (University of California, Irvine--UCI)

*Sinclair Lab: (Barnes Maze, Open Field Test (OFT), Tube dominance)) - (Harvard University)

*Soltesz Lab: (3D behavior signatures) - (Stanford)

Desai Lab: (Touchscreen based assays) - (Harvard University)

Electrophysiology:

*Wood Lab: Extracellular field recordings from the CA1 - (UCI)

*Soltesz Lab: Whole cell electrophysiology - (Stanford)

Neurotransmitter analyses in vivo:

*Desai Lab (Harvard University)

Ultrastructural studies on neurons: Electron microscopy, neuronal morphology

*Dickstein Lab (Uniformed Services University of Health Sciences)

Vestibular/Behavioral endpoints:

*Wallace Lab (Northern Illinois University)

Theory of Criticality: Neuronal connectivity

*Beggs lab (Indiana University)

Much of the foregoing work will be finalized by the end of 2021.

Lastly, we have made significant progress on examining the potential adverse effects of negative and positive pion exposure on cognition in mice. Targeting doses between 5-20 cGy using 150 MeV pions, we have significant cognitive deficits at 4 and 7 months post-exposure, suggesting these subatomic particles do pose a risk to astronauts for manifesting a certain level of cognitive dysfunction during deep space travel. We expect to have finalized these studies by the end of our funding period.

Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Krishnan B, Natarajan C, Bourne KZ, Alikhani L, Wang J, Sowa A, Groen K, Perry B, Dickstein DL, Baulch JE, Limoli CL, Britten RA. "Chronic low dose neutron exposure results in altered neurotransmission properties of the hippocampus-prefrontal cortex axis in both mice and rats." Int J Mol Sci. 2021 Apr;22(7):3668. https://doi.org/10.3390/ijms22073668 ; PMID: 33915974 , Apr-2021
Articles in Peer-reviewed Journals	Parihar VK, Angulo MC, Allen BD, Syage A, Usmani MT, Passerat de la Chapelle E, Amin AN, Flores L, Lin X, Giedzinski E, Limoli CL. "Sex-specific cognitive deficits following space radiation exposure." Front Behav Neurosci. 2020 Sep 16;14:535885. https://doi.org/10.3389/fnbeh.2020.535885 ; PMID: 33192361; PMCID: PMC7525092 , Sep-2020
Articles in Peer-reviewed Journals	Klein PM, Parihar VK, Szabo GG, Zöldi M, Angulo MC, Allen BD, Amin AN, Nguyen QA, Katona I, Baulch JE, Limoli CL, Soltesz I. "Detrimental impacts of mixed-ion radiation on nervous system function." Neurobiol Dis. 2021 Apr;151:105252. https://doi.org/10.1016/j.nbd.2021.105252 ; PMID: 33418069. , Apr-2021
Articles in Peer-reviewed Journals	Keiser AA, Kramár EA, Dong T, Shanur S, Pirodan M, Ru N, Acharya MM, Baulch JE, Limoli CL, Wood MA. "Systemic HDAC3 inhibition ameliorates impairments in synaptic plasticity caused by simulated galactic cosmic radiation exposure in male mice." Neurobiol Learn Mem. 2021 Feb;178:107367. https://doi.org/10.1016/j.nlm.2020.107367 ; PMID: 33359392 , Feb-2021
Articles in Peer-reviewed Journals	Britten RA, Wellman LL, Sanford LD. "Progressive increase in the complexity and translatability of rodent testing to assess space-radiation induced cognitive impairment." Neurosci Biobehav Rev. 2021 Jul;126:159-74. Available online 22 March 2021. Review. https://doi.org/10.1016/j.neubiorev.2021.01.027 ; PMID: 33766676 , Jul-2021
Articles in Peer-reviewed Journals	Willey JS, Britten RA, Blaber E, Tahimic CGT, Chancellor J, Mortreux M, Sanford LD, Kubik AJ, Delp MD, Mao XW. "The individual and combined effects of spaceflight radiation and microgravity on biologic systems and functional outcomes." J Environ Sci Health C Toxicol Carcinog. 2021 Apr 27;39(2):129-79. https://doi.org/10.1080/26896583.2021.1885283 ; PMID: 33902391 , Apr-2021